Additional file 1. Pre-registration

Pre-Registered Primary Endpoint

Project: BRCA-PathwayML v2 (revision after Communications Medicine + Scientific Reports desk rejections).

Target venue: npj Breast Cancer (primary), BMC Cancer (backup).

Primary claim: A non-linear machine learning model (Random Survival Forest or Gradient-Boosted Survival or DeepSurv, whichever wins on internal CV) trained on 7 interpretable biological pathway scores is non-inferior to PAM50 and the Oncotype DX 21-gene surrogate on overall discrimination AND is superior to both on the triple-negative (TNBC) subgroup, across at least three independent breast cancer cohorts.

Primary endpoint: TNBC meta-analyzed delta C-index (best ML model vs PAM50-ROR) across all cohorts with TNBC patients, random-effects meta-analysis.

Success threshold (PRE-REGISTERED): Meta-analyzed delta C-index >= +0.03 in TNBC, p < 0.05.

Secondary claims:
(a) ML model outperforms Cox baseline by >= +0.02 C-index on at least one cohort (showing real ML adds value).
(b) SHAP / model-class-appropriate attribution identifies immune-axis pathway as dominant in TNBC and HER2+, recovering known biological pattern.
(c) Combined model (ML pathway + PAM50 + clinical) shows IDI > 0.05 over PAM50 alone.
(d) ML model non-inferior to PAM50 (one-sided, margin -0.02) on overall cohort.

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Modifications after analysis begins: NOT PERMITTED.